

PP application: A task management tool for people with ADHD that amplifies cognitive advantages to correct disadvantages.

Using the six strengths of people with ADHDer to circumvent the six symptoms of their executive disorder, based on the neurodiversity and mindfulness model.

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Neurodiversity refers to how people's brains work and interpret information, and ADHD is a common form of neurodiversity. Characterised by age-inappropriate hyperactivity, impulsivity and inattention, it is defined as a cognitive deficit. However, less scientific literature has focused on its positive effects. Current digital applications that assist people with ADHD in managing tasks are also mostly based on control tools such as control, reminders and overcoming. In addition to their limited effectiveness, these tools reinforce the negative mental models of people with ADHD. Previous research has shown that adults with ADHD have a strong capacity for unconscious thought. However, at the same time, neurodiversity manifests itself in impaired execution, which is one of their core symptoms. In order to better avoid executive impairment in the lives of adults with ADHD, while building on their strengths, this paper offers a digital application that can be used to compensate for shortcomings based on the six positive effects of ADHD. Because "people with differences may not need a cure; they need help and accommodation".



CCS CONCEPTS • Human-centered computing • Human computer interaction (HCI) • Interactive systems and tools • User interface programming

Additional Keywords and Phrases: ADHD, Neurodiversity, Cognitive advantages, Digital application.

1 INTRODUCTION

Forcing a person with ADHD to pull back their attention is like taking away a nearsighted person's glasses. Behavioural characteristics of ADHD do not exist in a binary form of normal and abnormal [1]. A person with ADHD has deficits in many aspects of cognition, including cognitive control, sensitivity to rewards and time perception. At the same time, however, they have high energy and motivation, creativity, hyperfocus, easy-going, empathy and a willingness to help others. This means some aspects can adapt rather than be corrected [1]. The American Psychiatric Association defines the dual behavioural profile phenomenon of ADHD as inattention and hyperactivity-impulsivity; there are at least 13 ADHD symptoms, and this dual-processing model decreases in apparent ADHD as the child grows into adulthood, but problems with organisation and attention persist. The core cognitive deficit is executive dysfunction [2], which refers to a broad range of 'top-down' cognitive processes and abilities that enable flexible, goal-directed behaviour [3]. It is also one of the shortcomings in the social activities of the adult ADHD population. Current approaches to alleviating the executive symptoms of ADHD include traditional medications and behavioural therapies. However, they cannot maintain long-term symptom relief and have severe limitations in terms of side effects [4], so psychosocial interventions are a necessary complement to pharmacological interventions [5]. One of the most widely adapted psychosocial intervention tools is the digital app, which can assist patients in implementing strategies and enhancing skills. In a review of 355 previous studies of effective apps related to ADHD [6], 63 were related to executive management. In studies of the effectiveness of these apps, most were negative [6], such as poor outcomes not being maintained long after the intervention [7] and more severe anxiety when separated from the intelligent tool [8]. This also contributes to the mental model inherent in people with ADHD. Therefore the design assumptions in this paper are no longer based on limiting executive dysfunction in ADHD but instead move towards harnessing their cognitive motivation.

ADHD was the first neurodiversity symptom caused by a specific neurotransmitter deficiency. A fundamental assumption of the neurodiversity paradigm is that all forms of neurodiversity are valuable [9]. Brown, in 2005 developed a model that outlines the effects of ADHD on executive functioning in terms of six dimensions: activation, concentration, effort, emotion, memory and execution. On the other hand, in a qualitative study of the phenomenology of ADHD, six core themes of positive aspects were defined, namely cognitive vitality, courage, energy, humanity, resilience and transcendence [3]. In this paper, based on the neurodiversity paradigm of ADHD, the cognitive strengths element of ADHD is used as a core tool to design a program that enables unconscious states to circumvent executive disadvantage. An attempt is made to design a task management tool that can pull back the attention of ADHD without taking off their 'glasses'.

2 RELATED BACKGROUNDS

The neurodiversity of executive functions in ADHD is concentrated in four brain regions: the prefrontal cortex, the basal ganglia, the cerebellum, the parietal-temporal region and the reticular activating system [10, 11, 12, 13]. It has yet to be possible to categorise each brain region and the behavioural features it produces independently; they are interconnected and are a dynamic and integrated management process. However, executive processing can only function when the activities of multiple components of the cognitive structure are coordinated [14]. This chapter will analyse the diversity of executive functions in ADHD in combination, find possible connections in which positive and negative aspects can be interchanged through the neural systems that contribute most to its performance, and explore whether this can be used as a logical framework for a

mobile application module. Further, as none of the task management tools is designed to address the positive effects of ADHD, the only way to summarise and modify the following is by reviewing the functional boards of the applications related to the symptoms of their executive disorder. Finally, the principles and methods needed to develop a mobile application are reviewed. All three are combined for prototyping.

2.1 Research Overview

2.1.1 The Cognitive Vitality VS Concentration.

Cognitive vitality is the first positive core theme of ADHD and can correspond to the executive impairment of concentration; its conceptualisation summarises the uninterrupted mental activity of people with ADHD, spontaneous and non-sequential thought processes, flashes of images, and intense mental concentration. This context can be defined as a perception of an intensely pleasurable self-experience as an effective agent of change that, through its intervention, can rapidly alter the course of events [15]. The corresponding neural system is the brain's default network, negatively activated during external attention tasks and positively activated during internal psychological tasks related to the self, moral judgments, situational memories, and future visions [16]. In ADHD, the brain's default network is active, resulting in executive dysfunction in terms of the inability to focus and complete an external task for long periods. Neurodiversity is reflected in the inability to concentrate when faced with an external task and the inability to stop when the internal task requires a break.

So the cognitive dynamism of divergent thinking, creativity, curiosity and a quick knowledge base can be used as a management tool for design [17], correcting for the inability of attention to be drawn in.

2.1.2 The Courage VS Emotion.

As a second core theme, Courage can correspond to the implementation barrier of emotions; its conceptualisation summarises the process by which people with ADHD face their fears and deal with uncertainty. It can be defined as the struggle to overcome gaining recognition and striving for individualisation. The struggle for individualisation is a struggle to transcend restrictive social norms and beliefs through resistance to cultural adaptation or self-determination [18]. The neural system that responds to this is the amygdala, which is richly interconnected with several cortical areas in the thalamus and ventral striatum and plays a crucial role in emotion regulation and reward representation. The amygdala is often over-activated in patients with ADHD, including surface morphology and dopamine receptor density [19], while the ventral striatum is under-activated [20]. Correspondingly, this leads to executive dysfunction as restlessness, inadequate emotional self-regulation and impulsivity. Neurodiversity is reflected in low effect in the face of positive stimuli and over-perception in the face of negative stimuli [21, 22]; rapid decision-making is possible in the face of short-term rewards, but aversive delays occur in anticipation and acceptance of long-term rewards [23].

So the non-conformity, risk-taking, integrity and persistence in courage can be used as management tools designed to avoid procrastination; but not to avoid emotional outbursts.

2.1.3 The Energy VS Activation.

As a third theme, energy corresponds to the activation of executive disorders. Its conceptualisation summarises the psycho-energetic processes that generate positive internal drives in people with ADHD, which can be defined as willpower motivated by a strong desire to achieve something. This universal energy connects

people with ADHD to nature, associated with a guided subjective experience, a collective unconscious quest for higher aspects that encompass the self-aspect, and a sense of purpose and meaning in life [24]. The neural system that responds to this is the dorsal attention network (DAN) [25], which can be thought of as the camera's 'aperture' in the brain, responsible for focusing and directing attention to a salient stimulus, often in conjunction with the activation of the ventral attention network (VAN), which plays a more prominent role in shifting our attention. The VAN plays a more prominent role in shifting our attention.

In contrast, the DAN is more active in ADHD patients than the VAN, with the selective top-down allocation of attention being higher than due bottom-up attention [26]. The corresponding executive impairment is the inability to activate the temporal priority of the task. ADHD does not see time as a sequence but as a dispersed collection of events instinctively linked to the people, activities and emotions involved. This is manifested by the inability to organise and prioritise tasks correctly because attention is not easily shifted. Neurodiversity is reflected in the fact that people with ADHD are hyper focused on the task at hand when there is an ambitious goal in the ADHDer's ideology [20], but at the same time, neglect priority tasks due to events that are instinctively linked to them (Diamond, 2013).

So the willpower to achieve expectations and goals in energy can be used as a management tool designed to activate prioritised tasks and reorder the ADHD chaotic concept of time.

2.1.4 The Humanity VS Execution.

Humanity as a fourth theme corresponds to the execution of executive disorders; its conceptualization summarises the identification process of social intelligence, humour, self-acceptance and feelings of people with ADHD when engaging in social behaviour. It can be positioned as establishing management relationships through higher emotional intelligence [27], empathy, awareness and attitude [28]. The neural system that responds to this is the mirror neuron, which produces the same feelings as others when they are observed to produce emotional feelings. This leads to corresponding executive impairment as difficulty in changing one's behaviour. Neurodiversity is reflected in the tendency to emit positive energy to infect others when complex social events occur but, simultaneously, the inability to adapt one's behaviour to the situation when others are puzzled or angry.

Positive energy in human nature can be used as an extension tool in the design to maintain positive emotions in people with ADHD. Still, behaviour change requires other tools to intervene.

2.1.5 The Resilience VS Effort.

Resilience, as a fifth theme, corresponds to the effort of executive impairment; its conceptualization describes how people with ADHD choose to cope with ADHD. It can be defined as the process of self-regulation and self-protection under adverse conditions, which includes behavioural adjustment strategies for self-adherence [29]. The neural system that responds to this is the anterior cingulate cortex, located in the medial frontal lobe and plays an extensive role in cognitive control and behavioural monitoring [30]. The corresponding executive impairment that results is the inability to manage alertness. Neurodiversity is reflected in the activation

of the anterior cingulate cortex when antagonistic conflict arises, with people with ADHD exercising a degree of self-control but then ineffective self-regulation due to mental fatigue caused by ego depletion [31].

Self-regulation of behaviour in resilience cannot be used as a stand-alone management tool in the design. However, it must be added to other adjustment strategies to achieve effective management in choice.

2.1.6 The Transcendence VS Memory Loss.

Transcendence, as the final core theme, corresponds to the amnesia of executive disorder; its conceptualisation describes the sensory process of appreciation of beauty and excellence in people with ADHD, typically in the case of music. Appreciation of beauty and excellence is a new topic in positive psychology[2]. It can be defined here as the feeling of being quickly astonished by the talent in the environment, with associated feelings of awe, wonder, and memorisation. The neural system that responds to this is the corticocerebellar network with norepinephrine and the corticocerebellar network with internal functions such as the executive control network that generates links and is responsible for working memory. Whereas the internal links with norepinephrine are less activated in people with ADHD, neurodiversity is reflected in the fact that people with ADHD show high absorption and openness to experience when recalling experiences related to artistic beauty, for example [32], but are unable to recall stimulus- and stress-free details such as ordinary scenes.

Deep memories of appreciating artistic beauty in Transcendence can be designed with component tools to help deepen working memory.

2.2 Mobile Tools Research

2.2.1 User Interaction Design.

Mental models are the key to user-centred design [33]. A good mental model increases user\computer interaction and optimises performance. A mental model is a person's abstract view of a concrete event, and a mental model in design is a concept of the device that users have, providing them with the ability to interpret and predict that application [34]. In a survey on the relationship between mental models and reasoning, it was noted that graphical interfaces would produce higher performance than textual interpretations and that the output of a particular culture through visual, symbolic, and pictorial elements in design is the process of conceptualising mental models [35]. According to the physical symbol system hypothesis, any physical symbol can represent the simulated user's mental functions [36]. A good mental model can therefore tell the story of the design, represent the position of the application, and extend or even reshape the user's perception.

Regarding UI design principles, based on a systematic literature review of past UI design principles, six core principles were selected from the 257 most cited principles, in order: providing informative feedback, minimising the user's memory load, good error messages, providing shortcuts, speaking the user's language, and using real-world metaphors [37].

2.2.2 Interactive Modules.

This paper reviews 22 representative past management tools available for adults with ADHD (10,000-50,000 downloads), as shown in Table 1. The resulting application modules are shown in Table 2.

Table 1: Tool Resources

Tools Name	Categories
My ADHD	Diagnostic test
ADHD Test	Diagnostic test
Brain Focus	Tasks management
SimpleMind	Organize thoughts
Productive Habit Tracker	Build habits
Brili Routines	Build habits
RescueTime	Time management
Focus@Will	Focus and attention
Freedom	Tasks management
Mint	Financial management
Google Voice	Focus and attention
Boomerang for Gmail	Tasks management
Dropbox	Memory management
IFTTT	Tasks management
ScheduleOnce	Tasks management
Finish	Tasks management
WriteRoom	Time management
Brainsparker	Creation
Sleep as Android	Emotion
Dragon	Focus and attention
Priority Matrix	Time management
Due Reminders & Timers	Tasks management

Table 2: Tool References

Disfunction	Symptoms	Module	Element	Conference
Activate	Time chaos	Task reminder	Mandatory reminder	No
Activate	Time chaos	Label classification	Capture information	Yes
Focus	Lack of attention	Timer	Mandatory reminder	No
Hard work	Inability to keep	Timer	Mandatory reminder	No
Emotion	Impulsivedecision	Meditation	Mindful relaxation	Yes
Memory loss	Missing tasks	Mapping	Visual Spatialization	Yes
Execution	Inability to adjust behavior	Habit tracking, personalized setting time	Enhancing self-awareness	Yes

Module	Function	Advantages	Conference
Management roles	User feedback	Power	Yes
Communication	Patient Interaction	Humanity	Yes
Finance	Investment management	Power	Yes
Feedback	Express	Power	Yes
Random Creations	Dispersed thinking	Cognitive advantages	Yes

2.3 Research Conclusions

1. Cognitive dynamism can be adapted to focus disorders by using the creativity and divergent thinking bias of ADHD patients to draw their attention away from intrinsic tasks and towards the need for extrinsic tasks.
2. Negative stimuli cannot be used instead of positive stimuli to evoke rapid decision-making in ADHD patients. Because anxiety is unavoidable when faced with a long-term reward task, multiple timely reward tasks can be used instead of long-term reward tasks to evoke positive impulsive decision-making in ADHD patients.
3. Modify the concept of a priority task by replacing it with the process of achieving a higher goal; by making the subjective experience of a higher meaning its ultimate goal object, the person with ADHD is motivated to have the willpower that the higher goal must be achieved in order to complete the priority task.
4. Presenting tasks that are important and require working memory in an artistic form to reduce working memory loss.
5. Add a community section to the tool beyond the patient, using social connections with others to keep people with ADHD in a positive mood.
6. Modules that could be drawn upon are tagging categories, positive thinking, mapping, habit tracking, personalising set times, managing posts, user communication, financial management, calendars, and random ideas.
7. The inherent mental model of people with ADHD is mostly negative, such as 'I need to be managed'. In the design framework, the positivity of their mental model needs to be extended with positive elements, thus giving it a broader meaning and using a virtuous circle to motivate people with ADHD to take the initiative.
8. UI visuals should adhere to the six core principles above and the process of conceptualising the creation of a mental model.

Figure 2: Setting of delta threshold region

3 USER GROUP AND REQUIREMENTS

Previous studies have shown that 2.8% of adults worldwide have ADHD, with prevalence rates ranging from 1.2% to 4.4% [38]. In recent decades, diagnosis rates have dramatically increased in most countries [39].

Concerning the social needs of people with ADHD, the first is their need for external affirmation of positive aspects of themselves. In an analysis of two qualitative studies from 2020 and 2018 (422 adults with ADHD in six countries), only eight participants reported no positive aspects. The other 374 participants perceived a positive effect of ADHD on themselves but little social recognition. When combined with the research themes in this paper, it was concluded that people with ADHD described cognitive vitality such as problem-solving solid skills "rich in fantasy"; courage such as "always seeking novelty" and "an outsider to many of our benefits"; descriptions of energy such as "can focus on details for long periods" "can concentrate well if interest is stimulated". descriptions of humanity, such as "likes to find their back story" "has a strong sense of justice"; descriptions of resilience, e.g. "has to try to act completely different" "Got to find a balance". Descriptions of transcendence such as: "It was an excellent experience" "I will always remember the funny stories and pictures".

Although people with ADHD can recognise their strengths, there are still worries about self-esteem and social functioning. In a review of the literature on ADHD that included 127 studies [40], it was concluded that 29% and 56% of treated and untreated adults with ADHD, respectively, felt that their self-esteem was low, and 38% and 73% felt that their social functioning was affected, with living arrangements, activities and hobbies having a more significant impact.

So people with ADHD need social recognition of their perceived strengths on the one hand and want to address low executive skills and self-esteem on the other.

4 DESIGN PROCESS

Based on the discussion above, this is combined with user needs. The modular logic of the application will be based on the neurodiversity of ADHDers to improve their executive abilities in social functioning; the design will be based on extending the positive mental model of ADHDers to improve their sense of self-esteem. The available positive factors will be mapped to the needs of ADHD users, outlining the commonalities of their visual and interactive language, including modules, colours and visual effects. Temporarily name it PP (positive and power) application.

4.1 Module Logic

The logic of the application module is designed as follows. Using "Cognitive Vitality" to enter the concentration task section, the user can flip five favourite images and click on them to reveal the words behind them. The user can create their work based on these five random terms, but the subject of the creation is the external task they need to concentrate on. Before the puzzle, the user enters information about the task they need to remember. Each piece of information will randomly correspond to each small piece of the puzzle. When the user hovers over a piece of the puzzle and hesitates about its position, the information corresponding to that piece will always be displayed in the blank. Use the 'Habit Tracker Map Editor' to access the Persistent

Tasks section, a page where the user is the centre of the universe, and the other planets are all active events related to the user. Each completed will light up a corresponding planet. Use "Instant Rewards" to access the Long Term Tasks section, where the user breaks down the Long Term Tasks into Short Term Tasks, each corresponding to a reward. When users are full of positive emotions, they can use links to express themselves, such as helping other patients or user feedback; when negative emotions come, they can use positive thinking modules, such as meditation. When the user wants to be an actor, he/she can simulate the clothes and scenes from his/her favourite film or TV production to complete the priority tasks in reality. The logic is as in the figure 1.

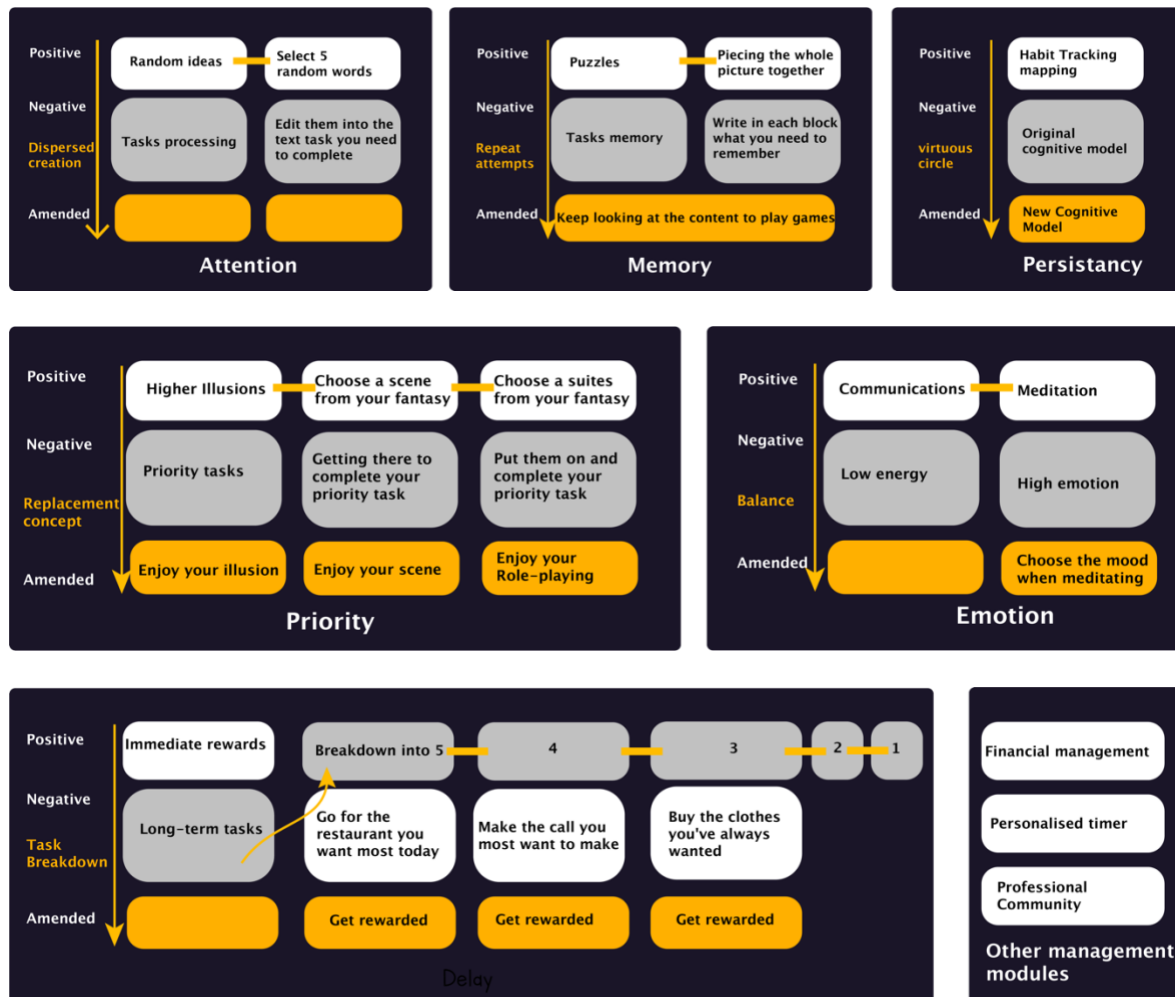


Figure 1: Module logic

4.2 Workflow Setting

As shown in figure 2.

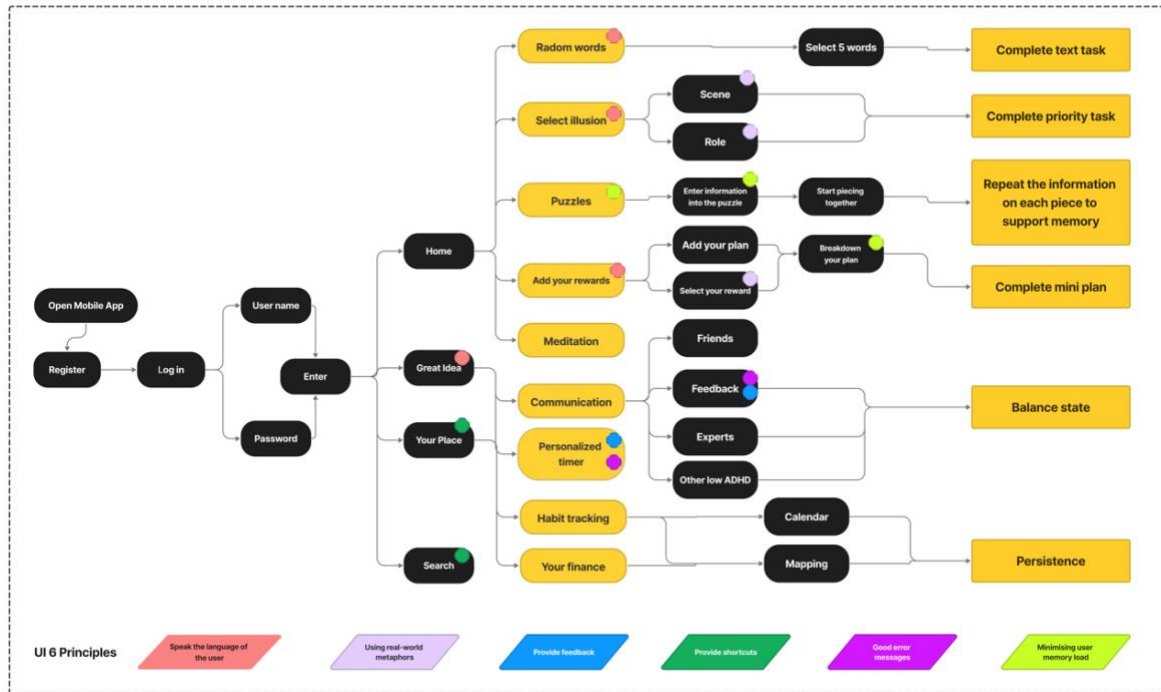


Figure 2: PP workflow

5 PROTOTYPE

5.1 Prototype Framework

As shown in figure 3.



Figure 3: PP Wireframe diagram

5.2 High Fidelity Setting

The visual effects aim to assist the positive module in addressing executive impairment and extending the mental model of people with ADHD. People with ADHD have more sensitive visuospatial processing skills. Therefore, design elements should be less repetitive and less patterned. Colours should be chosen to represent the positive aspects of their high energy level; icons should be chosen as irregular blocks that match their creativity and divergent thinking; fun puzzle sections should use unique artwork and main page visual displays such as figure..



Figure 4: PP application UI design

6 USER TESTING

User feedback was obtained after testing by three anonymous ADHD users, as shown in figure . Positive feedback included: the app is fun, and the colours attract attention. Negative feedback was that although it starts with interest, the noise still tends to distract users in an external environment (e.g. the underground). Resulting test results: PP should add music and sound effects to increase the user's sense of focus on the experience.

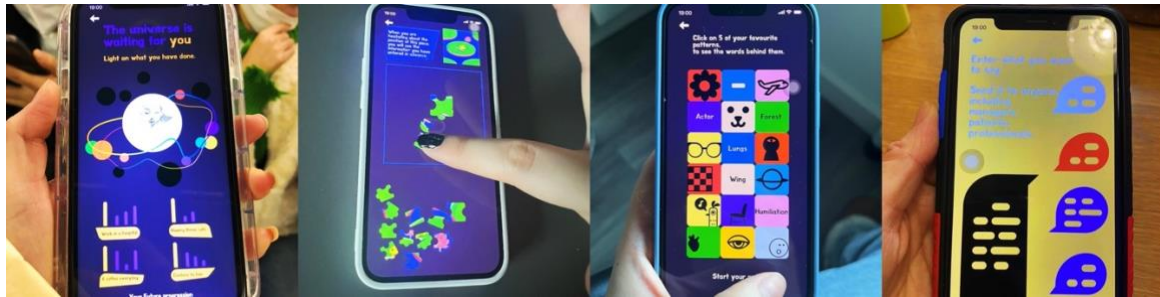


Figure 5: PP User testing

7 DISCUSSION

7.1 Main Findings

This paper provides, for the first time, a mobile application based on ADHD strengths to assist in circumventing their implementation disadvantages. It is a daily task management tool for people with ADHD that replaces coercion, reminders, corrections and control with creativity, high energy, dispersion and high output. The application module logic is based on the neurodiversity of people with ADHD. The interaction principles are based on the positive mental model of people with ADHD, and the visual design is based on the strengths bias of people with ADHD.

7.2 Limitation

Fewer applications and studies are available for people with ADHD who suffer from deafblindness, including the PP mobile application in this paper. An extensive test sample is also needed to validate the effectiveness of this tool that uses positive effects to circumvent deficits.

7.3 Iteration Work

Based on the results of user testing, PP will add music and sound effects as a next step; here, past research has shown that neurofeedback applications based on standard protocols are the well-researched and proven adjunctive treatment for ADHD.¹⁶ As the cost of wireless EEG devices gradually decreases, PP will add new treatments in the future to include neurofeedback technology. This approach is also applicable to patients with deaf blindness.

8 CONCLUSION

ADHD was often considered a cognitive defect in the past, and its cognitive advantages were usually ignored, causing ADHD patients to suffer from many inappropriate treatments in the workplace and in life. This design not only hopes to help the executive deficits of ADHD patients through their energy but also hopes that more people can change their cognitive models of ADHD after using this design.

9 ETHICAL STATEMENTS

Following local legislation and institutional requirements, ethical review and approval were not required for those who participated in the user testing. Participants all submitted their informed consent for this study via the UCL Redcap link.

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